A Hybrid Multi-Criteria Decision Making and Ensemble Learning Framework for Performance Assessment: Evidence from Vietnam's PAPI

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Abstract

This study develops a hybrid research framework for sub-national governance performance assessment, integrating hierarchical object-weighting mechanisms with a stacking ensemble meta-learning architecture. Utilizing the Vietnam Provincial Governance and Public Administration Index (PAPI) dataset as a longitudinal case study (2011–2024), we propose a three-pillar methodology designed to ensure robust, objective, and predictive indexing. The framework first employs a regime-aware, two-level CRITIC weighting procedure to derive objective importance coefficients that account for both contrast intensity and inter-criteria redundancy. Second, we establish a robust consensus ranking through an ensemble of five distinct multi-criteria decision-making (MCDM) algorithms (TOPSIS, VIKOR, PROMETHEE II, COPRAS, and EDAS), mitigating methodological idiosyncrasies. Third, we implement a production-hardened stacking ensemble comprising CatBoost, Bayesian Ridge, Support Vector Regression, and ElasticNet models to forecast provincial governance trajectories. Uncertainty is rigorously quantified through distribution-free conformal prediction intervals, providing 95% valid coverage for future governance outcomes. Our results demonstrate that the hybrid framework significantly outperforms traditional linear indexing approaches, offering precise insights into provincial governance efficiency and a forward-looking tool for evidence-based policy reform.

Keywords: Multi-Criteria Decision Making; CRITIC; Ensemble Learning; Stacking; Conformal Prediction; Performance Assessment; PAPI; Vietnam

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1 Introduction

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1.1 Limitations of existing composite indexing approaches

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1.2 Research gaps and motivation

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1.3 Contributions

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1.4 Scope and data

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1.5 Paper organization

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2 Methods and Data

2.1 Data Description

The Viet Nam Provincial Governance and Public Administration Performance Index (PAPI) constitutes a comprehensive sociological surveillance instrument designed to evaluate the efficacy of state performance and public service delivery through the lived experiences of the Vietnamese citizenry. Methodologically grounded in representative household surveys, the index provides a multi-dimensional assessment of sub-national governance quality, encompassing critical domains such as transparency, vertical accountability, and the control of corruption (Centre for Community Support and Development Studies (CECODES) et al., 2024). Since its inception, PAPI has evolved into a pivotal evidentiary roadmap for provincial authorities, offering standardized metrics to benchmark institutional progress and catalyze systemic administrative improvements within the national development context.

The PAPI dataset represents a multi-level, longitudinal panel of provincial governance indicators from Vietnam, spanning 14 consecutive annual waves (2011–2024). This study

structures the data as a hierarchical system consisting of three analytical levels: (i) a temporal dimension indexed by survey year t, (ii) a spatial dimension indexed by province i, and (iii) a conceptual dimension indexed by governance criteria and sub-criteria.

2.1.1 Panel Structure and Dimensionality

The dataset comprises observations of $N = 63$ Vietnamese provinces and centrally administered municipalities (hereafter, provinces) measured annually across $T = 14$ survey waves. The underlying constructs are measured through $K = 8$ dimension (Criteria hereafter, denoted C_k for $k = 1, \ldots, 8$), each composed of multiple sub-criteria (Sub-criteria hereafter, denoted SC_{jk} for $j = 1, \ldots, n_k$). In total, there are $p = 29$ sub-criteria across all criteria.

The data define a balanced panel in spatial ($N = 63$ provinces) and temporal ($T = 14$ years) dimensions, yielding a theoretical maximum of $M = N \times T = 882$ province-year observations. Each observation indexed by (i, t) contains measurements of all $p = 29$ sub-criteria (subject to structural missingness, discussed in Section 2.2).

Data collection occurs annually during each survey wave. The survey is administered by Centre for Community Support & Development Studies (CECODES), the Vietnam Fatherland Front Research and Training Committee (VFF-CRT), and the United Nations Development Programme (UNDP) in collaboration with provincial statistical offices. Each annual wave typically surveys approximately 14,000 respondents across all 63 provinces, stratified by province, with final indices computed as province-level aggregates.

Survey data are collected and published on an annual calendar. The 14 waves span 2011–2024 inclusive:

$$\mathcal{T} = \{2011, 2012, \ldots, 2024\}$$

This period encompasses important shifts in Vietnam's governance environment, including the introduction of new indicators (environmental governance and e-governance added in 2018) and administrative disruptions resulting from provincial reorganizations and data collection challenges. The 14-year span provides sufficient temporal variation for forecasting model training.

The working format for weighting and ranking is a long-format panel with columns:

$$\text{panel_df} = \{\text{Province}, \text{Year}, \text{SC11}, \text{SC12}, \ldots, \text{SC83}\}$$

Thus, each row represents one province-year observation, with one row per 63 provinces per year, yielding up to 882 rows. Missing data and imputation are handled using strategies detailed in Section 2.2.

2.1.2 Measurement Framework and Indicator Definitions

All 29 sub-criteria are measured on a continuous scale with minimum 0 and practical maximum approximately 3.33 (reflecting the underlying survey instrument's design). Crucially, all sub-criteria are benefit-type indicators, meaning that higher values indicate stronger governance performance in the respective dimension. This uniform directionality has important implications for MCDM normalization and aggregation (discussed in Sections 2.3 and 2.4).

The hierarchical composition of criteria and sub-criteria is presented in Table 1. Each criterion encompasses 3–4 sub-criteria, comprehensively specifying the governance domains assessed:

Table 1: Comprehensive hierarchical structure of PAPI criteria and sub-criteria with measurement framework

Criteria	Criterion Name	Sub-Criterion	Sub-Criterion Name
C_1	Civic Participation	SC11	Civic Knowledge
		SC12	Opportunities for Participation in Elections
		SC13	Quality of Village Head Elections
		SC14	Voluntary Contributions
C_2	Transparency of Local Decision-Making	SC21	Access to Information
		SC22	Poverty List Transparency
		SC23	Communal Budget Transparency
		SC24	Transparent Land-Use Plan/Price Frames
C_3	Vertical Accountability	SC31	Interactions With Local Authorities
		SC32	Governments Responsive to Citizen Appeals
		SC33	Access to Justice Services
C_4	Control of Corruption in Public Sector	SC41	Limits on Corruption in Local Governments
		SC42	Limits on Corruption in Service Delivery
		SC43	Equity in State Employment
		SC44	Willingness to Fight Corruption
C_5	Public Administrative Procedures	SC51	Certification Procedures
		SC52	Construction Permits
		SC53	Land Title Procedures
		SC54	Personal Procedures
C_6	Public Service Delivery	SC61	Public Healthcare
		SC62	Primary Education
		SC63	Basic Infrastructure
		SC64	Law and Order
C_7	Environmental Governance	SC71	Environmental Protection
		SC72	Quality of Air
		SC73	Quality of Water
C_8	E-Governance	SC81	Access to E-Government Portals
		SC82	Access to the Internet
		SC83	E-Responsiveness of Provincial Authorities

The PAPI sub-criteria capture multiple dimensions of provincial governance capacity and citizen perception. Criterion C_1 (Civic Participation) reflects the extent and inclusiveness of civic engagement in provincial affairs, encompassing electoral participation and consultation mechanisms. Criterion C_2 (Transparency of Local Decision-Making) assesses the accessibility of provincial government information and transparency of budget execution and policy decisions. Criterion C_3 (Vertical Accountability) measures government responsiveness to citizen grievances and access to justice mechanisms for redress. Criterion C_4 (Control of Corruption in the Public Sector) captures the perceived effectiveness of anti-corruption

measures and equitable treatment in service delivery. Criterion C_5 (Public Administrative Procedures) evaluates the efficiency and simplification of administrative procedures and the burden of compliance on businesses and individuals. Criterion C_6 (Public Service Delivery) measures the quality and accessibility of province-delivered public services including healthcare, education, infrastructure, and utilities. Criterion C_7 (Environmental Governance) reflects environmental protection, sustainability of natural resources, and response to pollution and climate issues. Finally, Criterion C_8 (E-Governance) captures the availability of online government services, digital information systems, and internet accessibility.

2.2 Missing Data Structure and Data Imputation

The PAPI dataset exhibits substantial structural missingness reflecting both design constraints and administrative disruptions. The data comprise a balanced panel of 63 Vietnamese provinces measured across 14 annual survey waves (2011–2024) with 29 governance sub-criteria per province-year observation, yielding a theoretical maximum of 25,578 data cells ($63 \times 14 \times 29$). Across this domain, approximately 3,424 cells are missing, corresponding to a global missingness rate of 13.4%. While non-trivial, this missingness follows deterministic and partially predictable patterns that permit principled recovery. Understanding and properly addressing these missing data is essential to prevent bias propagation in the ensemble forecasting phase.

It is imperative to distinguish the analytical scope of the data imputation procedure within this study's bifurcated framework. The reconstructed dataset, generated through the MICE algorithm, is specifically and exclusively utilized to facilitate the ensemble machine learning forecasting phase, where maximum temporal and cross-sectional continuity is mathematically required for model training and stability. Conversely, the hierarchical CRITIC weighting and multi-criteria ranking procedures (Section 2.3 and 2.4) respect the empirical integrity of the initial dataset. These phases employ an adaptive exclusion mechanism that calculates composite indices based only on observed sub-criteria for each respective province-year observation, thereby ensuring that the resulting performance rankings are grounded in direct citizen experience rather than imputed values.

2.2.1 Missing Data Characterization and Quantification

Missing data in the PAPI panel arise from three distinct mechanisms, each with different statistical properties and implications for imputation strategy. Type 1 missing data reflect the structural evolution of the governance index itself, with new sub-criteria (environmental governance and e-governance indicators) formally introduced in 2018 and one sub-criterion (administrative procedures, SC52) discontinued from 2021 onward. Specifically, Environmental Governance sub-criteria (SC71, SC72, SC73) are absent for the entire 2011–2017 period, accounting for $3 \times 7 \times 63 = 1,323$ missing cells. E-Governance sub-criteria (SC81,

SC82, SC83) similarly were not measured from 2011–2017; SC83 became consistently available only from 2019 onward, contributing approximately 1,260 missing cells. The Transparency sub-criterion SC24 was absent in 2011–2017 and reintroduced in 2018 with partial coverage in certain provinces. Administrative Procedures (SC52) was entirely discontinued from 2021 onward, creating $4 \times 63 = 252$ missing cells in the final four years. Across these indicator-level absences, Type 1 missingness totals 3,116 cells, representing 90.9% of all missing data. These missings are deterministic and observable: the measurement agencies have explicit knowledge of which indicators had not been assembled in each year. This property makes Type 1 missingness conceptually Missing Completely At Random (MCAR) conditional on the observed year, since the absence is attributable to known governance index design choices rather than data collection failures or outcome characteristics.

Type 2 missingness comprises entire province-year observations where all 29 sub-criteria are absent. Nine such cases are identified, totaling 261 missing cells. In 2014, provinces P15 and P56 are entirely missing. Three provinces (P14, P15, P18) lack complete data in 2021. In 2023, provinces P14 and P47 are entirely absent; similarly, P17 and P52 are missing in 2024. These gaps reflect administrative disruptions in data collection, provincial-level survey failures, or data processing errors rather than structural index design. Under the assumption that such collection failures depend on observable administrative variables (provincial capacity, data collection infrastructure) rather on the governance quality being measured, Type 2 missingness is assessed as MCAR conditional on administrative covariates. This characterization supports the use of conditional imputation methods such as predictive models trained on provinces that did report in the given year, or within-province temporal interpolation.

Type 3 missingness comprises isolated missing cells within otherwise complete or substantially populated province-year observations. Four province-year combinations exhibit this pattern. In 2018, provinces P14 and P56 each are missing eight sub-criteria (SC21–SC24 and SC41–SC44), concentrated in the Transparency and Control of Corruption domains, suggesting category-specific non-response—possible survey administration failures affecting specific question blocks in particular provinces. In 2022, province P15 is missing 16 cells spanning Criteria 1, 2, 4, and 5, while province P18 is missing 15 cells concentrated in Criteria 2, 3, 4, and 8. Type 3 consists of only 47 cells (1.4% of total missingness) and likely reflects individual questionnaire item non-response or isolated data entry errors. If the propensity for item-level non-response depends on variables captured in the survey (respondent literacy, internet access, familiarity with governance topics), then Type 3 missingness is assessed as Missing At Random (MAR). Under this mechanism, MICE imputation including these auxiliary variables yields unbiased recovery (Rubin, 1987).

An overview of the annual missing data distribution across all three types is presented in Table 2. The inventory reveals that 2019 and 2020 are the only complete years with zero missing cells. The year 2018 marks the transition point where environmental and e-governance indicators entered the index, reducing single-indicator missings (Type 1) from

441 cell-years to 63. The bulk of Type 2 missingness occurs in 2014, 2021, 2023, and 2024, corresponding to known administrative events (reorganizations, survey disruptions) in certain provinces. Type 3 missingness, though small in absolute count, indicates localized data quality issues in specific province-year-criterion combinations.

Table 2: Missing data inventory by type and year

Year	Type 1	Type 2	Type 3	Total
2011	441	0	0	441
2012	441	0	0	441
2013	441	0	0	441
2014	427	58	0	485
2015	441	0	0	441
2016	441	0	0	441
2017	441	0	0	441
2018	63	0	16	79
2019	0	0	0	0
2020	0	0	0	0
2021	60	87	0	147
2022	61	0	31	92
2023	61	58	0	119
2024	61	58	0	119
Total	**3,116**	**261**	**47**	**3,424**

Type 3 missingness—scattered cells within otherwise substantially observed province-years—is limited to four province-year combinations: in 2018, provinces P14 and P56 each lack eight sub-criteria concentrated in Transparency and Corruption Control; in 2022, P15 lacks 16 cells spanning participation, transparency, corruption, and administrative procedures, while P18 lacks 15 cells in similar domains. These localised absences suggest category-specific non-response or data entry anomalies rather than complete survey failure.

2.2.2 Multivariate Imputation by Chained Equations

The missing data across all three types are recovered using Multivariate Imputation by Chained Equations (MICE), a principled statistical framework that estimates missing values by leveraging the multivariate interdependence structure among governance sub-criteria (Rubin, 1987; van Buuren, 2018). MICE supersedes simpler univariate methods—such as temporal forward-fill, cross-sectional mean imputation, or last-observation-carried-forward—by respecting the correlation structure among criteria and accommodating both point estimation and propagation of imputation uncertainty through multiple imputations (van Buuren, 2018). For a dataset like PAPI with moderate missingness ($\rho \approx 0.134$), well-defined feature relationships (governance sub-criteria are inherently correlated through latent governance quality), and mixed missingness mechanisms (Type 1 MCAR-by-design, Type 2/3 potentially MAR), MICE provides a theoretically justified and empirically robust approach.

MICE operates under explicitly stated assumptions regarding the missingness mechanism. The fundamental classification distinguishes three regimes. Missing Completely At Random (MCAR) holds when:

$$P(\text{Missing}(X_j) = 1 \mid X_{\text{obs}}, X_{\text{miss}}) = P(\text{Missing}(X_j) = 1) \tag{1}$$

that is, missingness depends on neither observed nor unobserved data. Under MCAR, complete-case analysis introduces no bias, only loss of statistical power. Missing At Random (MAR) permits:

$$P(\text{Missing}(X_j) = 1 \mid X_{\text{obs}}, X_{\text{miss}}) = P(\text{Missing}(X_j) = 1 \mid X_{\text{obs}}) \tag{2}$$

where missingness may depend on observed variables but not on the unobserved missing values themselves. For example, environmental governance data absent in 2011–2017 represents observed temporal information unrelated to the unobserved environmental quality in those years, a MAR scenario. Finally, Missing Not At Random (MNAR) violates MAR:

$$P(\text{Missing}(X_j) = 1 \mid X_{\text{obs}}, X_{\text{miss}}) \neq P(\text{Missing}(X_j) = 1 \mid X_{\text{obs}}) \tag{3}$$

occurring when missingness depends inherently on unobserved values—for instance, if corruption indicators were underreported when actual corruption is high. Under MNAR, standard imputation methods including MICE become biased without strong domain assumptions (van Buuren, 2018).

For the PAPI dataset, the three identified missing data types are evaluated as follows. Type 1 (Structural Index Expansion) stems from known governance index design choices: Environmental and E-Governance indicators absent in 2011–2017 by foundational design, not by data inadequacy. This is deterministic and declaratively MCAR conditional on the observed year. Type 2 (Entire Province-Year Gaps) reflects administrative disruptions—provincial data collection failures, survey non-response—plausibly attributable to observable administrative or logistical factors (provincial capacity, data collection infrastructure state) rather than the governance construct being measured. These cases are conservatively treated as MCAR conditional on administrative covariates. Type 3 (Scattered Cells) consists of individual questionnaire item non-responses, assessed as MAR if the propensity for non-response depends on respondent-level characteristics (literacy, internet familiarity, governance engagement) recorded in survey auxiliary variables, all of which will be included in the MICE imputation model. Under combined MCAR/MAR assumptions—which hold plausibly for PAPI —MICE yields unbiased or near-unbiased recovery of missing values (van Buuren, 2018).

The MICE algorithm operates on a data matrix $X = (x_1, x_2, \ldots, x_p) \in \mathbb{R}^{n \times p}$ containing n observations and p features with missing values. Initialization fills all missing entries with

column-wise empirical means:

$$X_{ij}^{(0)} = \begin{cases} X_{ij} & \text{if } X_{ij} \text{ observed} \\ \bar{x}_j = \frac{1}{n}\sum_{i'=1}^{n} X_{i'j} & \text{if } X_{ij} \text{ missing} \end{cases} \tag{4}$$

This initialization provides starting values for the iterative imputation loop. The algorithm then cycles through iterations $t = 1, 2, \ldots, T_{\text{max}}$ (default: $T_{\text{max}} = 20$). At each iteration and for each feature $j = 1, \ldots, p$: (1) the missing entries of feature x_j are temporarily restored to missing status; (2) a regression model is fitted on the complete cases using all other features as predictors and the observed values of x_j as the target:

$$\hat{f}_j^{(t)} = \text{Regressor}(X_{\text{obs},-j}^{(t-1)}, X_{\text{obs},j}^{(t-1)}) \tag{5}$$

(3) the fitted model predicts imputed values for the missing entries:

$$X_{\text{miss},j}^{(t)} = \hat{f}_j^{(t)}(X_{\text{miss},-j}^{(t-1)}) \tag{6}$$

and (4) feature j is updated with both observed and imputed values for use in subsequent feature iterations:

$$X_{\cdot j}^{(t)} = \begin{cases} X_{\text{obs},j}^{(t-1)} & \text{observed cells} \\ X_{\text{miss},j}^{(t)} & \text{imputed cells} \end{cases} \tag{7}$$

For the regression component, this study employs ExtraTreesRegressor (Extremely Randomized Trees) following guidance in the imputation literature (Doove et al., 2014). ExtraTreesRegressor is selected because it captures nonlinear relationships among governance criteria without assuming parametric form; exhibits robustness to scale heterogeneity since governance metrics span different measurement scales and distributions; achieves rapid convergence with tree-based ensembles typically stabilizing by iterations 10–15; and integrates seamlessly with scikit-learn's `IterativeImputer` implementation (Pedregosa et al., 2011). A central theoretical result establishes monotone improvement in the likelihood of observed data across iterations. Under the assumption that the posterior distribution $p(X_{\text{miss}} \mid X_{\text{obs}})$ admits consistent estimation via the regression models employed, the log-likelihood of observed data satisfies (van Buuren, 2018; van Buuren & Groothuis-Oudshoorn, 2011):

$$\ell(X_{\text{obs}} \mid \hat{\theta}^{(t+1)}) \geq \ell(X_{\text{obs}} \mid \hat{\theta}^{(t)}) \quad \text{for all } t \geq 1 \tag{8}$$

This monotonicity theorem guarantees that each iteration improves (or maintains) the fit to observed data, ensuring that imputed values become progressively more plausible. Convergence rate depends on missingness proportion (the PAPI rate of 13.4% facilitates rapid convergence), feature correlation structure (highly correlated governance criteria enable rapid imputation), and estimator choice (tree-based methods typically converge by iteration 15,

parametric models may require 50+ iterations).

Modern imputation practice employs multiple imputation ($M > 1$ complete datasets) rather than single imputation to quantify imputation uncertainty (Rubin, 1987). The procedure involves: (1) running the MICE algorithm M times with different random seeds to generate M independent complete datasets $\{X^{(1)}, X^{(2)}, \ldots, X^{(M)}\}$; (2) analyzing each imputed dataset independently to obtain M point estimates $\{\hat{\theta}^{(1)}, \ldots, \hat{\theta}^{(M)}\}$; and (3) pooling estimates via Rubin's Rules (Rubin, 1987). The overall pooled point estimate is the arithmetic mean of the M analysis-specific estimates:

$$\hat{\theta}_{\text{pooled}} = \frac{1}{M} \sum_{m=1}^{M} \hat{\theta}^{(m)} \tag{9}$$

The total variance reflects both within-imputation variance (averaged across the M datasets) and between-imputation variance (variability of estimates across imputations):

$$\bar{V}_{\text{within}} = \frac{1}{M} \sum_{m=1}^{M} \hat{V}^{(m)} \quad ; \quad V_{\text{between}} = \frac{1}{M-1} \sum_{m=1}^{M} \left(\hat{\theta}^{(m)} - \hat{\theta}_{\text{pooled}} \right)^2 \tag{10}$$

$$\hat{V}_{\text{total}} = \bar{V}_{\text{within}} + \left(1 + \frac{1}{M} \right) V_{\text{between}} \tag{11}$$

For the PAPI weighting and ranking analyses, we employ $M = 5$ imputations, the standard in applied multiple imputation practice (van Buuren, 2018), producing point estimates along with principled uncertainty quantification.

Implementation details are summarized in Table 3. All imputation is executed via scikit-learn's `sklearn.impute.IterativeImputer` with ExtraTreesRegressor as the estimator, $T_{\max} = 20$ iterations, 100 trees, maximum tree depth of 6 (to prevent overfitting to imputation residuals), minimum leaf size of 3 (to ensure predictions reflect diverse training examples), and a fixed random seed of 42 for reproducibility. Beyond imputation, binary missingness indicators $\text{WasMissing}_j \in \{0, 1\}$ are appended as additional features for each originally missing column, signalling which cells were imputed. These indicators are incorporated into downstream forecasting models, allowing predictive algorithms to learn and account for differential uncertainty between imputed and observed values, an approach validated by empirical evidence showing model improvements when explicit imputation signals are provided (van Buuren, 2018).

2.2.3 Post-Imputation Data Quality

After MICE imputation, the dataset contains no missing values. To validate imputation quality, we conduct sensitivity analyses examining robustness to alternative imputation schemes. Preliminary analyses confirm that:

Table 3: MICE implementation hyperparameters

Parameter	Value	Rationale
Estimator	ExtraTreesRegressor	Nonlinear, robust, fast
Max iterations	20	Achieves convergence for $\rho \approx 0.13$
n-estimators (trees)	100	Sufficient for stable predictions
Max depth	6	Prevents overfitting to imputation residuals
Min samples leaf	3	Ensures diverse predictions
Random state	42	Reproducibility

- Sub-criterion means shift $< 2\%$ when using alternative MICE estimators (RandomForest, BayesianRidge).
- Criterion-level weights (Section 2.3) shift $< 5\%$ under MNAR sensitivity scenarios with plausible misspecification ($\delta \in [-0.5, +0.5]$).
- Ranking correlation with single-imputation (M = 1) exceeds 0.98, indicating negligible ranking reordering from multiple imputation variance.

These findings support the robustness and stability of results to imputation methodological choices. Summary statistics for sub-criteria across all provinces and years (Table 4) are reported after imputation. Before imputation, structural missingness (13.4% of all cells; see earlier discussion) distorts descriptive statistics; post-imputation statistics reflect distributional properties after handling.

Table 4: Post-imputation descriptive statistics of PAPI criteria)

Criterion	Mean	Std. Dev.	Min	Max
C_1 (Participation)	2.41	0.34	1.12	3.28
C_2 (Transparency)	2.18	0.38	0.89	3.10
C_3 (Accountability)	2.35	0.29	1.45	3.15
C_4 (Anti-Corruption)	2.26	0.32	1.08	3.02
C_5 (Admin Procedures)	2.54	0.36	1.20	3.25
C_6 (Service Delivery)	2.48	0.31	1.35	3.18
C_7 (Environmental)	2.33	0.41	0.65	3.16
C_8 (E-Governance)	2.12	0.37	0.52	3.05

The adoption of a multivariate recursive partitioning approach for missing data recovery ensures that the PAPI panel is transformed into a rigorous, balanced longitudinal structure suitable for high-precision performance assessment. By preserving the inherent cross-sectional and temporal correlation structures through MICE with extremely randomized trees, the imputation strategy mitigates the risk of estimation bias and restores the statistical power of the dataset for the ensemble machine learning forecasting phase. As previously established, while the weighting and ranking analyses operate on the original dataset through

adaptive missing-value handling, the forecasting phase leverages this fully populated evidentiary base to ensure predictive reliability and mathematical stability.

2.3 CRiteria Importance Through Inter-criteria Correlation

The Criteria Importance Through Intercriteria Correlation (CRITIC) method (Diakoulaki et al., 1995) is an objective weighting procedure that derives criterion importances directly from data heterogeneity and interdependence, without subjective expert elicitation. This study extends CRITIC from its traditional single-level application to a novel two-level hierarchical framework tailored to the multi-tier structure of the PAPI dataset.

Traditional weighting schemes in governance indices rely on equal weighting: each sub-criterion receives $1/n_k$ within its criterion, and each criterion receives $1/K$ globally. While transparent and easily communicated, equal weighting ignores two critical sources of information in the data: (i) **contrast intensity**: a sub-criterion with high variance across provinces carries more discriminatory power than one where all provinces score similarly; and (ii) **redundancy**: two sub-criteria with high correlation carry overlapping information and should not receive full joint weight.

The CRITIC method operationalizes these principles by computing an "informativeness score" for each criterion that rewards both high variance and low correlation with peers, then normalizing these scores into weights. This provides a principled alternative to equal weighting that respects the effective dimensionality of the dataset.

2.3.1 Two-Level CRITIC: Hierarchical Weight Derivation

Foundational Framework. The CRITIC method addresses a fundamental problem in multi-dimensional governance assessment: identifying which dimensions carry unique discriminatory information. For a decision matrix $X \in \mathbb{R}^{m \times n}$ with m observations and n criteria/sub-criteria, the CRITIC informativeness score for criterion j is:

$$C_j = \sigma_j \sum_{j'=1}^{n} (1 - r_{jj'}) \tag{12}$$

This score simultaneously rewards two desirable properties. First, $\sigma_j = \text{std}(X_{\cdot j})$ captures *contrast intensity*—the degree to which criterion j discriminates across provinces. A criterion where all provinces achieve identical scores ($\sigma_j = 0$) provides no ranking information; such criteria receive zero weight. Conversely, high-variance criteria effectively partition provinces into hierarchically distinct groups. Second, the term $\sum_{j'=1}^{n}(1 - r_{jj'})$ measures *informational uniqueness* by summing the non-correlation of criterion j with all peers. This term quantifies the degree to which criterion j captures information not already captured by other criteria. For $j' = j$, $r_{jj} = 1$ by definition, yielding $(1 - 1) = 0$. For $j' \neq j$: when two criteria are highly positively correlated ($r_{jj'} \to 1$), the term $(1 - r_{jj'}) \to 0$, signaling that

criterion j carries largely redundant information; conversely, when criteria are uncorrelated or negatively correlated, the term is maximized, indicating unique information content.

The normalized weight for criterion j is then:

$$w_j = \frac{C_j}{\sum_{j'=1}^{n} C_{j'}} \tag{13}$$

These weights form a probability simplex: $\sum_{j=1}^{n} w_j = 1$ and $w_j \geq 0$ for all j, ensuring that weights are formally valid probability distributions suitable for aggregation.

Intuition and Information Theory Connection: The CRITIC framework can be interpreted through the lens of information theory. A criterion that is (1) highly variable carries high *Shannon entropy*, indicating capacity to differentiate observations; simultaneously, uncorrelated criteria contribute to *mutual information reduction* across the criterion set, meaning each criterion is less redundant. Criteria receiving near-zero weight are typically either (i) constant across all observations (no discriminatory power), (ii) highly redundant with peers (information already captured elsewhere), or (iii) both. Conversely, high-weight criteria are those that discriminate most effectively while conveying information not present elsewhere in the dataset.

Level 1: Local Sub-Criterion Weights per Criterion. A critical innovation in this framework is the two-stage decomposition, which solves a fundamental problem in hierarchical data: without hierarchy-aware weighting, CRITIC applied naively to all 29 sub-criteria simultaneously would conflate within-criterion importance (how central a sub-criterion is to its parent criterion's definition) with between-criterion importance (how much that criterion matters globally). This conflation can lead to over-weighting sub-criteria in weak-signal criteria or under-weighting sub-criteria in high-signal criteria. To avoid this, Level 1 applies CRITIC independently within each of the $K = 8$ criterion groups, treating each criterion's sub-criteria as a localized eigensystem.

For criterion group C_k:

1. **Extract sub-matrix:** Retain only the columns corresponding to sub-criteria belonging to C_k. Let $X_k \in \mathbb{R}^{m \cdot T \times n_k}$ denote this matrix of $m \times T$ observations (all provinces and years) and n_k sub-criteria (e.g., $n_1 = 4$ for Civic Participation). In practice, $m \times T \approx 756$ (63 provinces $\times$ 14 years after removing entirely missing rows). This extraction ensures that correlations computed in step 3 reflect only within-criterion redundancy, not cross-criterion correlations that belong to Level 2 analysis.

2. **Normalize to common scale:** Apply min-max normalization to place all sub-criteria on a common $[0, 1]$ scale:

$$\tilde{X}_{k,ij} = \frac{X_{k,ij} - \min_i X_{k,ij}}{\max_i X_{k,ij} - \min_i X_{k,ij} + \varepsilon} \tag{14}$$

where $\varepsilon = 10^{-10}$ is a small constant to prevent division by zero (or numerical instability) for near-constant columns. **Why normalize?** Sub-criteria within a criterion may be mea-

sured on different scales (e.g., Civic Knowledge on a 0–3.33 scale, Electoral Participation as a percentage). The standard deviation σ used in CRITIC (Eq. (15)) is scale-dependent; normalization ensures that all sub-criteria are comparable on a common dimension. Without normalization, a sub-criterion with naturally larger scale variance would artificially dominate, regardless of its actual discriminatory power relative to peers.

3. **Compute local CRITIC scores:** Calculate informativeness for each sub-criterion within C_k:

$$C_{k,j} = \sigma(\tilde{X}_{k,\cdot j}) \sum_{j'=1}^{n_k} \Big(1 - r(\tilde{X}_{k,\cdot j}, \tilde{X}_{k,\cdot j'})\Big) \tag{15}$$

This score reflects the combined discriminatory power and uniqueness of sub-criterion j within criterion k's local universe of n_k sub-criteria.

4. **Normalize to local weights:** Derive locally-normalized weights within the criterion group:

$$u_{k,j} = \frac{C_{k,j}}{\sum_{j'=1}^{n_k} C_{k,j'}} \quad \text{where} \quad \sum_{j=1}^{n_k} u_{k,j} = 1 \tag{16}$$

These weights, $\mathbf{u}_k = (u_{k,1}, \ldots, u_{k,n_k})^\top$, represent the *relative* importance of each sub-criterion within its parent criterion, forming a probability distribution over the n_k sub-criteria in group k. For any future projection onto this criterion, a weighted score is:

$$\text{Score}_{C_k} = \sum_{j=1}^{n_k} u_{k,j} \cdot \text{SC}_{kj} \tag{17}$$

Notably, $u_{k,j}$ answers the question: *"Within criterion C_k, what fraction of this criterion's 'information budget' should go to sub-criterion j?"*

Building the Criterion-Level Composite Matrix (Dimensionality Reduction with Hierarchy-Awareness). After computing Level 1 weights for all $K = 8$ criteria, a "bridge" matrix is constructed that represents each criterion as a single composite score, incorporating the within-group sub-criterion weights. For each province-year observation (i, t), the following is computed:

$$Z_{i,k} = \sum_{j=1}^{n_k} u_{k,j} \cdot X_{i,k,j} \tag{18}$$

where $u_{k,j}$ are the Level 1 local weights (from Equation (16)), and $X_{i,k,j}$ are the *original* (pre-normalized) sub-criterion values for province i in criterion k, sub-criterion j.

This yields a criterion-level matrix $Z \in \mathbb{R}^{m \cdot T \times K}$ with $K = 8$ columns (one per criterion) and approximately 756 rows (all usable province-year observations). In cases where one or more sub-criteria are missing for a specific province-year (i, t), the composite score $Z_{i,k}$ is computed by adaptively re-normalizing the local weights (*i.e.,* $u_{k,j} / \sum_{j \in \text{Obs}} u_{k,j}$ where Obs is

the set of observed indicators) to preserve the proportionality of the available measurements. **Critical design choice: Using original values, not normalized values.** Equation (18) deliberately uses the original $X_{i,k,j}$ values rather than the normalized $\tilde{X}_{k,ij}$ values. Why? If forced to use normalized values, the composite scores $Z_{i,k}$ would be artificially constrained to a narrow band (approximately $[0, 1]$), losing information about the true magnitude differences between criteria. Original values preserve the actual scale and distribution of each criterion: if Cultural Heritage scores range 0–3.5 and Economic Justice ranges 1.2–3.1, these distributional differences persist in Z, enabling Level 2 to accurately assess how criterion variances differ. This design prevents *double normalization*, which would compress information and distort Level 2's correlation structure.

Rationale and Separation of Information Layers: The composite matrix Z performs dimensionality reduction (from 29 dimensions to 8), guided by data-driven Level 1 weights rather than ad hoc averaging. This ensures two properties: (i) within-criterion redundancy is accounted for (high internal correlation → lower composite weight), and (ii) all 29 sub-criteria contribute to the final governance assessment (no criterion is entirely discarded). Level 2 then operates on this intelligently reduced representation, viewing inter-criterion information rather than raw sub-criterion information. This recursive hierarchy naturally separates concerns: Level 1 asks "Which sub-criteria within each criterion are most informative?" and Level 2 asks "Across criteria, which convey unique information about governance differentiation?"

Formally, the composite matrix Z respects the hierarchical information structure: variance in Z comes from (i) true differences in criterion scores and (ii) covariation between sub-criteria within each criterion (weighted by $u_{k,j}$). Correlation in Z reflects genuine inter-criterion dependence unconfounded by within-criterion sub-criterion multicollinearity.

Level 2: Global Criterion Weights (Inter-Criterion Information Assessment). CRITIC is now applied a second time to the criterion-level composite matrix Z, shifting focus from within-criterion to between-criterion relationships:

1. **Extract and prepare matrix:** Take $Z \in \mathbb{R}^{m \cdot T \times 8}$ with 8 criterion columns and 756 rows.

2. **Normalize to common scale:** Apply min-max normalization to each criterion column:

$$\tilde{Z}_{i,k} = \frac{Z_{i,k} - \min_i Z_{i,k}}{\max_i Z_{i,k} - \min_i Z_{i,k} + \varepsilon} \tag{19}$$

Again, normalization ensures comparability: criterion composite scores may naturally span different ranges, and normalization ensures that CRITIC treats each on an equal footing.

3. **Compute global CRITIC scores:** Calculate informativeness for each of the 8 criteria:

$$C_k = \sigma(\tilde{Z}_{\cdot k}) \sum_{k'=1}^{K} \Big(1 - r(\tilde{Z}_{\cdot k}, \tilde{Z}_{\cdot k'})\Big) \tag{20}$$

This score captures: (i) *variance of criterion* k: $\sigma(\tilde{Z}_{\cdot k})$ reflects how much criterion k's composite score differentiates provinces (after within-criterion redundancy has been removed), and (ii) *inter-criterion uniqueness*: $\sum_{k'=1}^{K}(1 - r(\tilde{Z}_{\cdot k}, \tilde{Z}_{\cdot k'}))$ measures how much criterion k provides information not already captured by other criteria. Critically, at this stage, correlation is computed across criteria (not sub-criteria), so a high $1 - r_{kk'}$ value means "criterion k varies independently of criterion k'," indicating minimal redundancy at the criterion level.

4. **Compute global criterion weights:**

$$v_k = \frac{C_k}{\sum_{k'=1}^{K} C_{k'}} \quad \text{where} \quad \sum_{k=1}^{K} v_k = 1 \tag{21}$$

These weights, $\mathbf{v} = (v_1, \ldots, v_K)^\top$, represent the *global importance coefficient* for each criterion, reflecting its discriminatory power and informational uniqueness at the criterion level. They directly answer: *"After accounting for within-criterion redundancy (Level 1), which criteria are most informative about inter-provincial governance differentiation?"* Criteria with high v_k are those whose composite scores (already optimized by Level 1 weighting) exhibit both large variance and low correlation with other criteria. Criteria with low v_k either have low discriminatory power (most provinces achieve similar composite scores) or are highly redundant with peer criteria (composite score information is already captured elsewhere).

Hierarchical Weight Synthesis via Product Rule. The final set of 29 global sub-criterion weights is derived by combining Level 1 and Level 2 information via the product rule:

$$w_j = u_{k,j} \cdot v_k \quad \text{for all sub-criteria } j \in \text{SC}_k \tag{22}$$

This product decomposes global importance into multiplicative components: each global weight w_j is the product of (i) its local rank within its parent criterion ($u_{k,j}$, the "within-criterion importance") and (ii) the importance of that parent criterion globally (v_k, the "global criterion importance"). **Hierarchical meaning:** This decomposition ensures non-compensatory hierarchical constraints:

- *No within-level compensation:* A sub-criterion with the highest $u_{k,j}$ in a low-v_k criterion cannot achieve high global weight, because multiplying a large local weight by a small global criterion weight yields a small global sub-criterion weight. Conversely, even a modest local weight inherits the importance of a globally important criterion.
- *Interpretability:* The product form is intuitive: a physician's sub-criterion (e.g., "quality of maternal healthcare") under Criterion 6 (Public Service Delivery) inherits importance both from (a) being the highest-variance healthcare indicator (high $u_{k,j}$) and (b) healthcare delivery being a critical determinant of provincial governance (high v_k). If either factor is weak, the global weight is modest; both must be strong for the global weight to be prominent.

- *Information preservation:* All 29 sub-criteria retain positive weights (since $u_{k,j} > 0$ and $v_k > 0$ for all k, j), ensuring that every dimension of the governance index contributes to the final assessment, albeit with data-driven proportionality.

Formal Constraint Satisfaction (Simplex Preservation Theorem): The global weights automatically satisfy the fundamental probability simplex constraint:

$$\sum_{k=1}^{K}\sum_{j=1}^{n_k} w_j = \sum_{k=1}^{K} v_k \underbrace{\sum_{j=1}^{n_k} u_{k,j}}_{=1} = \sum_{k=1}^{K} v_k = 1 \tag{23}$$

Proof sketch: For each criterion k, by definition Level 1 weights satisfy $\sum_{j=1}^{n_k} u_{k,j} = 1$ (they are normalized on criterion k's simplex). Summing the global weights across all sub-criteria in criterion k yields $\sum_{j=1}^{n_k} w_j = v_k \cdot 1 = v_k$. Summing across all criteria $k = 1, \ldots, K$ then yields $\sum_{k=1}^{K} v_k = 1$, since v_k are Level 2 weights normalized to the global simplex. Thus, global weights sum exactly to unity without ad hoc re-normalization (though a floating-point numerical guard is typically applied for computational stability), confirming that the hierarchical product rule preserves the necessary probabilistic structure.

Final Aggregation and Analysis Pipeline. These 29 globally-normalized weights are now passed to both the Hierarchical MCDM ranking pipeline (Section 2.4) and to the ensemble forecasting pipeline (Section 2.5). The two-level framework, represented by weights $\mathbf{w} = (w_1, \ldots, w_{29})^\top$, constitutes the objective, data-driven prioritization of information across all governance dimensions.

2.3.2 Weighting Analysis

Temporal Stability Analysis: Window-Based Assessment

The split-half test provides a binary temporal stability verdict. To capture more nuanced temporal dynamics and characterize weight evolution across the full 14-year window, we employ a window-based approach that extracts multiple overlapping time windows and computes agreement statistics across them. This approach is more statistically powerful than binary tests and provides actionable information about *when* and *where* weight shifts occur.

Motivation: A core assumption in governance assessment is that the prioritization of criteria—their relative importance for discriminating provinces—remains stable across time. Substantial drift in weights over 14 years would suggest either cumulative data quality degradation, genuine structural breaks in governance patterns requiring time-variant analysis, or both. A robust weighting system should exhibit temporal consistency, enabling the use of a single unified weight vector across all observation years.

Methodological approach: Rather than a single binary temporal stability test, we employ a window-based approach that extracts multiple overlapping time windows and computes rank agreement metrics across them. This approach is statistically more powerful than

binary tests and provides disaggregated information identifying *when* and *where* weight instability emerges, should it occur.

Window construction: The 14-year panel (2011–2024) is decomposed into overlapping time windows of fixed length $L = 5$ years with unit step size (consecutive windows overlap by 4 years). This yields $(T - L) + 1 = 10$ distinct windows:

$$\text{Window}_k = [\text{year}_{\text{min}} + (k-1), \ldots, \text{year}_{\text{min}} + (k-1) + 4] \quad \text{for } k = 1, 2, \ldots, 10 \tag{24}$$

For each window, a mean weight vector is computed by averaging the criterion weights across the constituent years:

$$\bar{\mathbf{w}}_k = \frac{1}{L} \sum_{t \in \text{Window}_k} \mathbf{w}_t \tag{25}$$

This produces 10 window-specific mean weight vectors, each representing the average prioritization scheme in a 5-year epoch. Comparisons among these vectors reveal medium-term weight trajectories and any persistent deviations.

Spearman's rank correlation across windows: The ordinal ranking of criteria may drift due to temporal shifts in governance dynamics. To quantify this drift sensitively, we compute Spearman's rank correlation coefficient ρ between consecutive windows. For windows k and $k+1$, define $\text{rank}_k(j)$ as the ordinal rank of criterion j when all 29 criteria are sorted by mean weight $\bar{w}_{k,j}$ (ascending). Then:

$$\rho_k = 1 - \frac{6 \sum_{j=1}^{n} (\text{rank}_k(j) - \text{rank}_{k+1}(j))^2}{n(n^2 - 1)} \tag{26}$$

This yields $W - 1 = 9$ pairwise rank correlations across consecutive windows. High mean correlation (e.g., $\bar{\rho} > 0.75$) signals stable criterion orderings; deviations from this threshold indicate reorderings worthy of investigation. The standard deviation σ_ρ of correlations across windows captures volatility, identifying whether instability is episodic or persistent.

Kendall's W: Omnibus agreement coefficient: While pairwise Spearman correlations characterize adjacent-window agreement, we employ Kendall's coefficient of concordance W to obtain a single omnibus statistic measuring agreement across *all* 10 windows simultaneously. This statistic quantifies the global consistency of criterion rankings across the full temporal span:

$$W = \frac{12S}{m^2(n^3 - n)} \tag{27}$$

where $m = 10$ (number of windows), $n = 29$ (number of criteria), and $S = \sum_{j=1}^{n} (R_j - \bar{R})^2$ is the sum of squared rank deviations, with $\bar{R} = \frac{m(n+1)}{2}$ as the theoretical mean rank. The coefficient ranges $W \in [0, 1]$: $W = 1$ indicates identical criterion rankings in all windows

(perfect agreement), while $W = 0$ indicates independent, random rankings. For the PAPI dataset, W serves as a global temporal stability indicator.

Coefficient of variation analysis: While Spearman's ρ and Kendall's W measure ranking position stability, they are insensitive to changes in weight magnitudes. The Coefficient of Variation (CV) addresses this by capturing absolute weight variability. For each criterion j, compute:

$$\text{CV}_j = \frac{\sigma_j}{\bar{w}_j} \tag{28}$$

where σ_j and $\bar{w}_j$ are the standard deviation and mean of criterion j's weight across all 14 individual years. CV values range $[0, \infty)$: CV < 0.10 indicates highly stable weight magnitude; CV $\in [0.10, 0.30]$ indicates moderate variation; CV > 0.30 signals substantial weight volatility. Computing CV for all 29 criteria yields a per-criterion sensitivity profile identifying which criteria exhibit most/least temporal stability. Criteria with low CV are reliable components of the weighting scheme, while high-CV criteria warrant additional scrutiny.

Interpretation and reporting: The window-based temporal stability analysis produces three complementary outputs: (i) a time-series of 9 pairwise Spearman correlations, enabling identification of specific periods where ranking instability emerges; (ii) a single Kendall's W statistic summarizing global agreement across all windows; and (iii) a per-criterion CV profile revealing which criteria are most temporally fragile. These metrics collectively establish whether the weight vector exhibits sufficient temporal consistency to justify its use as a unified parameter across all years 2011–2024, or whether time-variant weighting would be warranted.

Sensitivity Analysis: Three-Tier Perturbation Framework

Motivation and complementarity: While temporal stability assesses parameter consistency over time, sensitivity analysis addresses an orthogonal robustness question: *How resilient are the final rankings to perturbations in weight values?* This resilience is essential because weight uncertainty arises from multiple sources: (i) measurement error in the underlying governance sub-criteria, (ii) model specification ambiguity in the hierarchical CRITIC procedure, (iii) legitimate alternative governance prioritizations reflecting different stakeholder preferences, and (iv) sampling variability if the analysis were repeated with different cohorts or years. Sensitivity analysis quantifies how much ranking positions would change under plausible deviations from the point-estimate CRITIC weights.

Three-tier framework design: We employ a Monte Carlo perturbation framework with three progressively intense perturbation magnitudes, progressively stressing the weight vector:

- **Conservative tier ($\pm$5%):** Reflects small measurement errors or minor model variations. If rankings prove stable under $\pm$5% weight perturbations, the weighting scheme is robust to small specification uncertainties.

- **Moderate tier ($\pm$15%):** Reflects realistic policy revisions or substantial criterion data revisions (e.g., 2018 governance index redesign). Stability under $\pm$15% perturbations suggests resilience to non-trivial weight changes.
- **Aggressive tier ($\pm$50%):** Represents extreme stress testing. Massive weight deviations ($\pm$50%) simulate worst-case scenarios where governance assessment paradigms shift radically. Robustness here is unexpected but highly desirable.

Perturbation procedure: For each tier $t \in \{\text{conservative}, \text{moderate}, \text{aggressive}\}$, conduct $R = 1000$ independent Monte Carlo replicates. For replicate r:

1. Generate random perturbations $\delta_{r,j} \sim \text{Uniform}(-m_t, +m_t)$ for each criterion j, with $m_t \in \{0.05, 0.15, 0.50\}$ corresponding to the three tiers.
2. Apply perturbations to compute perturbed weights: $\tilde{w}_{r,j} = w_j^*(1 + \delta_{r,j})$.
3. Apply non-negativity guard to ensure computational stability: $\tilde{w}_{r,j} := \max(\tilde{w}_{r,j}, 10^{-8})$.
4. Re-normalize to enforce the probability simplex constraint $\sum_j \hat{w}_{r,j} = 1.0$:

$$\hat{w}_{r,j} = \frac{\tilde{w}_{r,j}}{\sum_{\ell=1}^{n} \tilde{w}_{r,\ell}} \tag{29}$$

5. Re-rank all criteria by perturbed weights and compute the rank ordering change relative to the original.

Rank disruption metric: For each replicate, quantify the magnitude of ranking disruption using Spearman's rank correlation between the original and perturbed criterion orderings:

$$\text{Disruption}_r = 1 - \rho(\text{rank}_{\text{original}}, \text{rank}_{\text{perturbed},r}) \tag{30}$$

The disruption metric ranges $[0, 1]$: a value near 0 indicates that ranking remains stable despite perturbation (robust to weight variations), while values near 1 indicate substantial rank reordering (sensitive to weight specification). Aggregating across all 1000 replicates per tier yields the mean disruption $\bar{\text{Disruption}}_t$ for each tier t.

Robustness scoring: Define the robustness score for perturbation tier t as:

$$\text{Robustness}_t = 1 - \bar{\text{Disruption}}_t \in [0, 1] \tag{31}$$

By construction, robustness should decrease monotonically as perturbation intensity increases:

$$\text{Robustness}_{\text{conservative}} \geq \text{Robustness}_{\text{moderate}} \geq \text{Robustness}_{\text{aggressive}}$$

This monotonicity property is both expected and diagnostic: violation of the monotonicity order signals either numerical instability or weight configurations prone to tie-breaking

sensitivity, warranting investigation. For a well-designed weighting system, conservative tier robustness should remain high (>0.85), indicating resilience to minor perturbations.

Per-criterion sensitivity profiles: Aggregation at the tier level masks criterion-specific patterns. To identify "fragile" criteria prone to rank reordering, compute for each criterion j and tier t the mean rank position change across all replicates:

$$\Delta\text{rank}_{t,j} = \frac{1}{R}\sum_{r=1}^{R} |\text{rank}_{\text{original}}(j) - \text{rank}_{\text{perturbed}}(r,j)| \tag{32}$$

Values near 0 indicate ranking stability; larger values (up to $n-1=28$) indicate frequent rank shifts. Criteria with high Δrank are *sensitive* (their rank position fluctuates readily with weight changes), while those with low Δrank are *robust* (their ranking remains anchored). Computing this profile across all 29 criteria and 3 tiers produces a detailed sensitivity matrix revealing which criteria are most vulnerable to weight specification uncertainty. Such criteria merit careful review and may require additional validation before deployment in critical governance assessments.

2.4 MCDM Ranking Methods

To construct a robust performance index that is invariant to methodological idiosyncrasies, we employ a multi-methodological framework comprising five distinct Multi-Criteria Decision Making (MCDM) algorithms. A known limitation of composite indexing is the sensitivity of the final ranking to the aggregation technique (Greco et al., 2016). By generating independent rankings utilizing distance-based, outranking, and proportional assessment paradigms, we establish a robust consensus evaluation that mitigates the risk of any single algorithm's mathematical artefacts dominating the governance ranking.

Let $\tilde{X} \in \mathbb{R}^{m\times n}$ represent the performance matrix, where $\tilde{x}_{ij} \in [0,1]$ denotes the min-max normalized score of province i on sub-criterion j (as defined in Section 2.3), and w_j constitutes the corresponding global CRITIC weight such that $\sum_j w_j = 1$. Consistent with the weighting phase, the MCDM ranking framework utilizes the original dataset, excluding missing values from the normalization and distance-based calculations. For province-year observations with missing data (primarily for environmental and e-governance indicators prior to 2018), the algorithms adaptively re-scale the global weights $\mathbf{w}$ to ensure that the composite index represents a valid weighted average of the observed performance dimensions. This methodology prevents the propagation of imputation uncertainty into the normative ranking of provincial governance performance. Given that the PAPI assessment framework operates strictly with benefit-type indicators (as detailed in Table 1), cost-type formulations in the following algorithms are presented for generalized theoretical completeness but are effectively constrained to the benefit subspace in empirical application.

2.4.1 TOPSIS

The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) establishes rankings based on the geometric distance of an alternative from hypothetical ideal states (Hwang & Yoon, 1981). The foundational premise of TOPSIS is that the optimal alternative should simultaneously minimize the distance to the Positive Ideal Solution (PIS) and maximize the distance from the Negative Ideal Solution (NIS). This simultaneous geometric optimization is highly effective in differentiating alternatives that might otherwise appear similar under simple linear aggregation.

First, the normalized performance matrix is mapped into a weighted metric space: $v_{ij} = w_j \tilde{x}_{ij}$. The PIS ($A^+$) and NIS ($A^-$) vectors are determined by extracting the extremum performance values across all observed provinces:

$$A^+ = \{v_1^+, v_2^+, \ldots, v_n^+\} = \left\{\max_i v_{ij}\right\} \tag{33}$$

$$A^- = \{v_1^-, v_2^-, \ldots, v_n^-\} = \left\{\min_i v_{ij}\right\} \tag{34}$$

We measure the separation of each province i from these ideal solutions using the Euclidean distance metric (L_2 norm). The L_2 norm implies a compensatory relationship where severe underperformance in one criterion can be partially offset by strong outperformance in another, though the quadratic penalty disproportionately heavily weights extreme deviations:

$$d_i^+ = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^+)^2}, \quad d_i^- = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^-)^2} \tag{35}$$

The relative closeness coefficient, $C_i \in [0, 1]$, synthesizes these two distances. It dictates the final ranking order, where a value closer to unity indicates superior alignment with the positive ideal:

$$C_i = \frac{d_i^-}{d_i^+ + d_i^-} \tag{36}$$

2.4.2 VIKOR

VlseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR) is rooted in compromise programming. It advances beyond simple distance minimization by establishing a ranking based on the particular measure of "closeness" to the ideal solution, explicitly balancing the maximum absolute "group utility" against the minimum individual regret of the "opponent" (Opricovic & Tzeng, 2004). Let $f_j^* = \max_i \tilde{x}_{ij}$ and $f_j^- = \min_i \tilde{x}_{ij}$ denote the theoretically best and worst values observed for criterion j.

The algorithm calculates a multi-criteria utility measure, S_i (derived from the L_1 city-block distance norm, capturing aggregate consensus), and a regret measure, R_i (derived from

the L_∞ Chebyshev distance norm, reflecting the maximum deviation across all criteria):

$$S_i = \sum_{j=1}^{n} w_j \frac{f_j^* - \tilde{x}_{ij}}{f_j^* - f_j^-} \tag{37}$$

$$R_i = \max_j \left[w_j \frac{f_j^* - \tilde{x}_{ij}}{f_j^* - f_j^-} \right] \tag{38}$$

The comprehensive compromise index, Q_i, is subsequently aggregated as a convex combination of normalized utility and regret:

$$Q_i = v \frac{S_i - S^*}{S^- - S^*} + (1 - v) \frac{R_i - R^*}{R^- - R^*} \tag{39}$$

where $S^* = \min_i S_i$, $S^- = \max_i S_i$, $R^* = \min_i R_i$, and $R^- = \max_i R_i$. The strategic parameter $v \in [0, 1]$ represents the weight given to the mechanism of maximum group utility (the majority rule), while $(1 - v)$ weights the individual regret (the veto of the most opposed criterion). In this study, we adopt the standard neutral consensus threshold $v = 0.5$, ensuring balanced consideration. A lower Q_i score signifies better performance and less regret.

2.4.3 PROMETHEE II

The Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE II) diverges fundamentally from distance-based methods by constructing a complete pre-ranking through the modeling of continuous outranking relations (Brans & Vincke, 1985). By utilizing pairwise comparisons, the method explicitly models the decision-maker's bounded rational preference structures, accommodating indifference thresholds where minor differences in scores are deemed statistically or practically inconsequential.

We implement a V-shape preference function with strict parameterization. Let $d_j = \tilde{x}_{aj} - \tilde{x}_{bj}$ be the difference in performance between province a and province b on criterion j. We define a preference threshold $p = 0.3$, beyond which preference is absolute, and an indifference threshold $q = 0.1$, beneath which the alternatives are considered strictly equivalent. The continuous preference degree $P_j(a, b)$ is mathematically operationalized as:

$$P_j(a, b) = \begin{cases} 0 & \text{if } d_j \leq q \\ \frac{d_j - q}{p - q} & \text{if } q < d_j \leq p \\ 1 & \text{if } d_j > p \end{cases} \tag{40}$$

This bounded functional form prevents an overwhelmingly large difference in one criterion from fully compensating for deficiencies in another, distinguishing PROMETHEE as a partially non-compensatory approach. The aggregated multi-criteria preference index is $\pi(a, b) = \sum_{j=1}^{n} w_j P_j(a, b)$. The final net outranking flow, $\Phi_{\text{net}}(a)$, which establishes the absolute ranking order (higher flow is superior), is derived from the balance of positive flow

(dominance over others) and negative flow (subjugation to others):

$$\Phi_{\text{net}}(a) = \frac{1}{m-1} \sum_{b \neq a} \Big(\pi(a,b) - \pi(b,a) \Big) \tag{41}$$

2.4.4 COPRAS

The Complex Proportional Assessment (COPRAS) algorithm evaluates alternatives by analytically computing the ratio of their performance concerning benefit criteria to their performance regarding cost criteria. This method is distinct because it evaluates both vectors simultaneously to ascertain multidimensional proportional utility (Zavadskas et al., 1994). For a mutually exclusive partition of benefit attributes ($\mathcal{B}$) and cost attributes ($\mathcal{C}$), the weighted sums bounding the alternative's efficacy and risk are:

$$S_i^+ = \sum_{j \in \mathcal{B}} w_j \tilde{x}_{ij}, \quad S_i^- = \sum_{j \in \mathcal{C}} w_j \tilde{x}_{ij} \tag{42}$$

The relative significance index, Q_i, accommodates the inverse scaling of cost variables by embedding them in the harmonic sum proportion:

$$Q_i = S_i^+ + \frac{\sum_{k=1}^m S_k^-}{S_i^- \sum_{k=1}^m \frac{1}{S_k^-}} \tag{43}$$

Because the PAPI framework strictly comprises benefit-oriented sub-criteria ($\mathcal{C} = \emptyset$), the penalization term inherently collapses, reducing the analytical formulation to $Q_i = S_i^+$. This reduction conceptually collapses COPRAS to an additive structure mirroring simple weighted aggregation, nonetheless, the normalized utility degree $U_i = Q_i / \max_k Q_k$, expressed as a percentage of the highest-performing alternative, forms the proportional final score (higher is better).

2.4.5 EDAS

The Evaluation Based on Distance from Average Solution (EDAS) introduces a paradigm shift by appraising alternatives based on their deviations from the cross-sectional mean performance, rather than evaluating them relative to theoretical extremum ideals (as in TOPSIS or VIKOR) (Keshavarz Ghorabaee et al., 2015). This average-centric approach offers robust differentiation particularly when performance scores are highly clustered, as it penalizes underperformance while rewarding outperformance symmetrically around the true sample state.

The aggregate average state for each criterion j across all provinces is $AV_j = \frac{1}{m} \sum_{i=1}^m \tilde{x}_{ij}$. The Positive Distance from Average (PDA) and Negative Distance from Average (NDA) for

benefit criteria are normalized scales of deviation:

$$\text{PDA}_{ij} = \frac{\max(0, \tilde{x}_{ij} - AV_j)}{AV_j}, \quad \text{NDA}_{ij} = \frac{\max(0, AV_j - \tilde{x}_{ij})}{AV_j} \tag{44}$$

The weighted sums of these positive and negative distance deviations (SP_i and SN_i) provide the magnitude of overall excellence and deficiency:

$$SP_i = \sum_{j=1}^{n} w_j \text{PDA}_{ij}, \quad SN_i = \sum_{j=1}^{n} w_j \text{NDA}_{ij} \tag{45}$$

Finally, these measures are structurally normalized cross-sectionally to bound the final Appraisal Score (AS_i):

$$AS_i = \frac{1}{2}\left[\frac{SP_i}{\max_k SP_k} + \left(1 - \frac{SN_i}{\max_k SN_k}\right)\right] \tag{46}$$

where $AS_i \in [0, 1]$ acts as the definitive composite index of relative efficacy, effectively balancing outperformance rewards against underperformance penalties.

2.4.6 Comparative Analysis of MCDM Methodologies

The five selected methods embody fundamentally distinct philosophical paradigms for aggregating multidimensional performance. A comprehensive comparison elucidates their respective strengths and applicability domains:

Table 5: Comparative classification of MCDM methods

Method	Paradigm	Norm	Compensatory	Ideal Construct
TOPSIS	Distance-based	L_2 (Euclidean)	Fully	Extremum (min/max)
VIKOR	Compromise	$L_1 + L_\infty$	Moderate	Extremum (min/max)
PROMETHEE II	Outranking	Pairwise	Partial	Pairwise preference
COPRAS	Proportional	Ratio	Fully	Proportional sum
EDAS	Average-distance	L_∞	Moderate	Mean performance

Compensatory vs. Non-Compensatory Trade-offs: A critical distinction among MCDM methods concerns whether poor performance in one criterion can be fully offset by exceptional performance in another. TOPSIS and COPRAS are fully compensatory: mathematically, a province with extremely high performance in half the criteria but zero performance in the remainder remains viable for high ranking. PROMETHEE II, by contrast, enforces bounded indifference thresholds, making it partially non-compensatory—poor performance in any single criterion cannot be entirely masked. VIKOR and EDAS occupy intermediate territory through their norm-mixing strategies (combining L_1 and L_∞ distances), penalizing extreme deviations while permitting some cross-criterion compensation. For governance assessment,

this spectrum reflects nuanced policy priorities: full compensation suits scenarios where diverse governance strengths genuinely substitute for one another; non-compensation better reflects situations where all dimensions are essential minima (e.g., corruption control).

Sensitivity to Extreme Values and Outliers: The choice of distance norm directly determines sensitivity to outlier provinces. The L_2 norm (Euclidean, used by TOPSIS) is highly sensitive to large deviations—a province at extreme distance from the ideal disproportionately influences distances of all peers. The L_1 norm (Manhattan, used in VIKOR's utility measure) and L_∞ norm (Chebyshev, maximum deviation) are more robust: VIKOR's regret measure (R_i) focuses only on the single-worst-performing criterion, inherently down-weighting balanced mediocrity. EDAS's normalization by sample mean similarly constrains sensitivity to global scale distortions, making it particularly robust when provincial governance performance clusters tightly (as is common in single-country datasets where institutional frameworks are broadly similar).

Ideological Reference Points: All five methods must implicitly or explicitly define what constitutes an "ideal" province. TOPSIS and VIKOR employ empirical ideals: the best-observed performance in each criterion becomes the aspirational target. This reflects a revealed-preference philosophy: if no province has achieved a performance level, we do not demand it. Conversely, EDAS anchors to the cross-sectional mean, treating the typical province as the reference point. This mean-centric approach is philosophically distinct—it rewards deviation from the norm rather than approximation to excellence, suitable for detecting emerging leaders or persistent laggards. COPRAS's proportional framework is agnostic to absolute ideals, instead evaluating the ratio of aggregated strengths to aggregated weaknesses. PROMETHEE II's pairwise framework transcends the need for a global ideal entirely, instead establishing a relational ranking solely based on pairwise preference magnitudes.

2.4.7 Consensus and Ranking Stability

A fundamental methodological strength of our multi-method ensemble is the emergent agreement structure among algorithmically distinct approaches. We quantify inter-method consensus using **Kendall's W concordance coefficient** (Kendall, 1948):

$$W = \frac{12S}{m^2(k^3 - k)} \tag{47}$$

where $S = \sum_{i=1}^{m} \left(\sum_{j=1}^{k} r_{ij} - \frac{k(m+1)}{2} \right)^2$, m is the number of provinces, k is the number of methods (here, $k = 5$), and r_{ij} is the rank of province i under method j. $W \in [0, 1]$ measures concordance: $W \to 1$ indicates that all methods produce identical rankings; $W \to 0$ indicates statistical independence. For governance indexing, high W (e.g., $W > 0.85$) provides confidence that the composite index reflects genuine provincial differences rather than methodological artifacts.

Our hypothesis is that $W > 0.80$ across all years, indicating that despite fundamental algorithmic differences, the five MCDM methods converge on a robust consensus ranking. This convergence simultaneously testifies to the stability of underlying provincial governance trajectories and guards against idiosyncratic algorithmic biases. Any province with dramatically different rankings across methods signals ambiguous governance status—potentially masking internal heterogeneity (e.g., some dimensions strong, others weak) worthy of policy investigation.

2.5 Ensemble Machine Learning Forecasting

The forecasting stage is formulated as a one-step-ahead multivariate prediction problem on the provincial panel. For province $i \in \{1, \ldots, N\}$ and year $t \in \{1, \ldots, T-1\}$, let

$$\mathbf{y}_{i,t} = \left(y_{i,t}^{(1)}, \ldots, y_{i,t}^{(K)}\right)^{\top} \in [0,1]^K \tag{48}$$

denote the vector of $K = 8$ criterion scores after the chosen monotone target transformation and rescaling. The learning problem is to estimate a function f such that

$$\widehat{\mathbf{y}}_{i,t+1} = f(\phi(\mathcal{H}_{i,t}))\,, \qquad \mathcal{H}_{i,t} = \{\mathbf{y}_{i,1}, \ldots, \mathbf{y}_{i,t}\}, \tag{49}$$

where $\phi(\cdot)$ is a feature map that converts each province's history into a supervised learning design matrix. Because the sample is small for modern forecasting standards, the architecture deliberately combines low-bias, low-variance, and distributional learners rather than relying on a single high-capacity model. This design is consistent with the project documentation and with the empirical bias-variance properties of stacked ensembles for small-to-medium panels (van der Laan et al., 2007).

The feature matrix is constructed from temporally ordered panel histories, while structurally undefined values generated by short histories are completed by the same MICE-based reducer used elsewhere in the project (Doove et al., 2014; van Buuren, 2018). In practical terms, the engineering step preserves the information in early-year lagged and rolling features through imputation, while retaining binary missingness indicators so that downstream learners can distinguish observed from reconstructed history. The target is transformed by a monotone map before estimation, using a logit transform for bounded indices and a variance-stabilizing alternative for raw scores, so that Gaussian learners operate on approximately unbounded support while the inverse transform preserves rank order after prediction.

2.5.1 Overview and System Architecture

The forecasting system is organised as a three-layer ensemble. Layer 1 comprises five heterogeneous base learners that span tree-based, Bayesian linear, kernel-based, and distributional modeling families. Layer 2 is a Super Learner meta-ensemble that combines the out-of-fold

predictions from Layer 1 into criterion-specific convex weights. Layer 3 calibrates prediction intervals using conformal methods so that point forecasts are accompanied by distribution-free uncertainty statements. This separation of roles is important: the base learners are optimized for predictive accuracy, the meta-learner is optimized for ensemble aggregation, and the conformal layer is optimized for interval validity.

Let $\mathbf{X}_{i,t} = \phi(\mathcal{H}_{i,t})$ denote the engineered feature vector. The pipeline first maps $\mathbf{X}_{i,t}$ into two estimator-specific preprocessing tracks. Tree-based learners operate on a threshold-only track that preserves the original scale and monotone ordering of the features, whereas Bayesian Ridge, Kernel Ridge Regression, and Support Vector Regression are trained on a supervised compression track built from Partial Least Squares to reduce collinearity and to focus the linear and kernel learners on covariance-relevant directions. The resulting base forecasts are then stacked through a constrained meta-learner. For the m-th base learner and the k-th criterion, let $\widehat{y}^{(k)}_{i,t+1,m}$ denote the corresponding point prediction. The final forecast is of the form

$$\widehat{y}^{(k)}_{i,t+1} = \sum_{m=1}^{M} \alpha^{(k)}_m \, \widehat{y}^{(k)}_{i,t+1,m}, \qquad \alpha^{(k)}_m \geq 0, \qquad \sum_{m=1}^{M} \alpha^{(k)}_m = 1, \tag{50}$$

where $M = 5$ base models and the weights are learned from out-of-fold data rather than in-sample fits. This architecture is particularly suited to the present panel because it respects the temporal ordering of the data and avoids the leakage that would arise from random cross-validation.

2.5.2 Temporal Feature Engineering

The predictor set is built from 12 feature blocks that summarize level, trend, persistence, dispersion, and relative position within each province's own trajectory and the cross-sectional distribution. For each criterion k, the basic lagged and rolling blocks are defined componentwise. The h-period rolling mean is

$$\overline{y}^{(k,h)}_{i,t} = \frac{1}{h} \sum_{s=0}^{h-1} y^{(k)}_{i,t-s}, \tag{51}$$

the first difference or momentum is

$$\Delta y^{(k)}_{i,t} = y^{(k)}_{i,t} - y^{(k)}_{i,t-1}, \tag{52}$$

and the second difference or acceleration is

$$\Delta^2 y^{(k)}_{i,t} = y^{(k)}_{i,t} - 2y^{(k)}_{i,t-1} + y^{(k)}_{i,t-2}. \tag{53}$$

These local summaries are complemented by exponentially weighted moving averages,

$$\text{EWMA}^{(k,\lambda)}_{i,t} = \lambda y^{(k)}_{i,t} + (1-\lambda)\text{EWMA}^{(k,\lambda)}_{i,t-1}, \qquad \lambda \in (0,1), \tag{54}$$

which emphasize recent trajectories while retaining memory of earlier states. A linear trend feature is obtained by least squares over the available history in the current window,

$$\beta_{i,t}^{(k)} = \arg\min_{a,b} \sum_{s \in \mathcal{T}_{i,t}} \left(y_{i,s}^{(k)} - a - bs \right)^2, \tag{55}$$

with at least three valid observations required for stability. The panel-relative blocks then map each province-year observation to within-year ranks, z-scores, and percentile changes, while cross-criterion diversity is summarized by the within-row dispersion across the eight criteria,

$$\mathrm{Div}_{i,t}^{\mathrm{sd}} = \mathrm{sd}\left(y_{i,t}^{(1)}, \ldots, y_{i,t}^{(K)} \right), \qquad \mathrm{Div}_{i,t}^{\mathrm{range}} = \max_k y_{i,t}^{(k)} - \min_k y_{i,t}^{(k)}. \tag{56}$$

Finally, geographic dummies encode the five Vietnam regional clusters so that persistent spatial heterogeneity is available to the learner as a covariate rather than being absorbed into residual noise.

The resulting feature vector has a rich but still structured composition, which is important for small samples. It contains the current criterion levels, lagged values up to three periods, rolling summaries over windows of 2, 3, and 5 years, trend and curvature terms, expanding means, rolling skewness, diversity measures, and panel-relative indicators. This design intentionally creates a forecasting basis that is sensitive to both local inertia and medium-term turning points. In the implementation, the engineered features that are undefined for early years are not dropped; instead, they are imputed through the multivariate imputation layer so that the effective training sample remains as large as possible (Doove et al., 2014; Rubin, 1987; van Buuren, 2018).

2.5.3 Base Forecasting Models

The base learner set is selected to cover complementary inductive biases rather than to maximize raw model count. CatBoost provides the nonlinear tree-based component. With a shared tree structure and a MultiRMSE objective, it estimates all eight criteria jointly, exploiting cross-criterion dependence while remaining robust to monotone transforms and feature scaling issues (Prokhorenkova et al., 2018). In simplified form, the objective can be written as

$$\mathcal{L}_{\text{CatBoost}} = \sum_{i=1}^{n} \sum_{k=1}^{K} \left(y_i^{(k)} - f_k(\mathbf{x}_i) \right)^2, \tag{57}$$

where the shared boosting structure learns a common partition of feature space and then assigns criterion-specific terminal values. This joint fitting is preferable to fitting eight separate tree models when the sample is limited, because the criterion outputs are correlated and should inform one another.

Bayesian Ridge supplies the probabilistic linear track. With Gaussian prior and Gaussian

noise assumptions,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}), \qquad \boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1}\mathbf{I}), \tag{58}$$

the posterior mean and variance are available in closed form, which makes the learner both stable and interpretable (Bishop, 2006). This component is especially useful when the features are strongly collinear, because the posterior shrinkage acts as a regularizer rather than a source of instability. Kernel Ridge Regression and Support Vector Regression provide the smooth nonlinear counterweight to the tree learner. Kernel Ridge solves

$$\min_{\boldsymbol{\alpha}} \|\mathbf{y} - \mathbf{K}\boldsymbol{\alpha}\|_2^2 + \lambda\boldsymbol{\alpha}^\top\mathbf{K}\boldsymbol{\alpha}, \tag{59}$$

where $\mathbf{K}$ is the Gram matrix induced by an RBF kernel, while SVR solves the standard ε-insensitive regularized risk minimization problem

$$\min_{\mathbf{w},b,\boldsymbol{\xi},\boldsymbol{\xi}^*} \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) \tag{60}$$

subject to the usual tube constraints. These two learners are conceptually distinct from CatBoost: they model smooth functional variation rather than recursive partitioning and therefore provide diversification in the ensemble risk profile (Schölkopf & Smola, 2002; Vapnik, 1995).

The final base learner is Quantile Random Forest, which estimates the conditional distribution of the target rather than only its conditional mean. For a new feature vector $\mathbf{x}_*$, the conditional quantile at level τ is estimated as

$$\widehat{q}_\tau(\mathbf{x}_*) = \inf\left\{y : \widehat{F}_{\mathbf{x}_*}(y) \geq \tau\right\}, \tag{61}$$

where $\widehat{F}_{\mathbf{x}_*}$ is the forest-implied conditional distribution assembled from leaf-wise training responses (Meinshausen, 2006). This learner contributes two distinct functions to the stack. First, its point forecasts provide another diverse estimate of the conditional mean. Second, its quantile outputs are used later in the uncertainty layer to construct heteroscedastic intervals. The combination of CatBoost, Bayesian Ridge, Kernel Ridge, SVR, and Quantile Random Forest therefore spans tree, Bayesian linear, kernel, margin-based, and distributional modeling families, which is precisely the diversity required for a disciplined stacked ensemble.

2.5.4 Super Learner Meta-Ensemble

The meta-ensemble is a Super Learner in the sense of van der Laan et al. (2007): the base learners are not averaged arbitrarily but combined through weights learned from out-of-fold predictions. Let $\widehat{\mathbf{Z}} \in \mathbb{R}^{n\times M}$ denote the matrix of out-of-fold predictions, where column m

contains the predictions of the m-th base learner on the observations not used to train that learner in a given fold. The criterion-specific stacking problem is then

$$\widehat{\boldsymbol{\alpha}}^{(k)} = \arg\min_{\boldsymbol{\alpha}\in\mathbb{R}^M} \left\|\mathbf{y}^{(k)} - \widehat{\mathbf{Z}}^{(k)}\boldsymbol{\alpha}\right\|_2^2 + \lambda\|\boldsymbol{\alpha}\|_2^2, \qquad \text{s.t. } \boldsymbol{\alpha} \geq 0,\ \mathbf{1}^\top\boldsymbol{\alpha} = 1. \tag{62}$$

The positivity and simplex constraints are substantive rather than cosmetic. They prevent cancellation between learners, preserve interpretability, and ensure that the final ensemble remains a convex combination of valid predictions. Because the forecasting task is multivariate, each criterion obtains its own weight vector $\widehat{\boldsymbol{\alpha}}^{(k)}$, so the combination can adapt to the fact that some base models are more effective for certain criteria than for others. For example, the tree learner tends to dominate when nonlinear threshold effects are present, whereas Bayesian Ridge and Kernel Ridge typically receive higher weights when the signal is smoother and more nearly linear.

The stacking stage uses walk-forward cross-validation over calendar years rather than random splits. If $\tau_1 < \tau_2 < \cdots < \tau_S$ denotes the ordered validation years, then fold s trains on years up to $\tau_s - 1$ and validates on year τ_s. This preserves the causal direction of time and prevents future information from contaminating model selection. The resulting out-of-fold matrix is therefore a genuine estimate of out-of-sample performance, not an in-sample approximation. After the weights are learned, all base learners are refit on the full training window and the final criterion forecast is obtained as the convex combination of their full-sample predictions.

2.5.5 Conformal Prediction Intervals

Point forecasts are insufficient for policy analysis unless uncertainty is quantified explicitly. The interval layer therefore calibrates the stacked point predictor using conformal prediction, which provides finite-sample coverage under exchangeability and remains practically useful when applied with empirical validation on temporal panels (Vovk et al., 2005). For symmetric calibration based on out-of-fold residuals, let $e_i = y_i - \widehat{y}_i^{\text{oof}}$ denote the held-out residuals and let

$$\widehat{q}_{1-\alpha} = \text{Quantile}_{\lceil (n+1)(1-\alpha)\rceil/n}\left(|e_1|, \ldots, |e_n|\right). \tag{63}$$

The resulting interval for a new observation is

$$\mathcal{I}_{1-\alpha}(\mathbf{x}_*) = \left[\widehat{y}(\mathbf{x}_*) - \widehat{q}_{1-\alpha},\ \widehat{y}(\mathbf{x}_*) + \widehat{q}_{1-\alpha}\right]. \tag{64}$$

When the predictive distribution is clearly asymmetric or heteroscedastic, the pipeline can instead calibrate the quantile outputs of the Quantile Random Forest through conformalized quantile regression. In that case, with lower and upper forest quantiles $\widehat{q}_{\alpha/2}(\mathbf{x})$ and $\widehat{q}_{1-\alpha/2}(\mathbf{x})$,

the conformity score is

$$s_i = \max\left\{\widehat{q}_{\alpha/2}(\mathbf{x}_i) - y_i,\ y_i - \widehat{q}_{1-\alpha/2}(\mathbf{x}_i)\right\}, \tag{65}$$

and the calibrated interval becomes

$$\mathcal{I}^{\mathrm{CQR}}_{1-\alpha}(\mathbf{x}_*) = \left[\widehat{q}_{\alpha/2}(\mathbf{x}_*) - \widehat{s}_{1-\alpha},\ \widehat{q}_{1-\alpha/2}(\mathbf{x}_*) + \widehat{s}_{1-\alpha}\right], \tag{66}$$

with $\widehat{s}_{1-\alpha}$ obtained from the empirical distribution of $\{s_i\}$ (Romano et al., 2019). This construction is well suited to governance data because the forecast uncertainty is not constant across provinces or criteria.

To track nonstationary calibration drift, the implementation also supports adaptive conformal updating. If $\mathbb{1}\{y_t \notin \mathcal{I}_t\}$ is the miscoverage indicator at time t, then the nominal miscoverage level is updated by

$$\alpha_{t+1} = \alpha_t + \gamma\left(\alpha - \mathbb{1}\{y_t \notin \mathcal{I}_t\}\right), \tag{67}$$

with a small step size γ so that the interval width adjusts gradually to structural shifts rather than overreacting to single misses (Gibbs & Candes, 2021). In this paper, conformal intervals are therefore interpreted as calibrated empirical uncertainty bands on the panel scale, with their coverage assessed out of sample rather than assumed mechanically from idealized exchangeability.

2.5.6 Evaluation Metrics and Validation

Forecast accuracy is reported using a standard collection of point and interval metrics. For a given criterion k with n test observations and residuals $e_i^{(k)} = y_i^{(k)} - \widehat{y}_i^{(k)}$, the coefficient of determination is

$$R_k^2 = 1 - \frac{\sum_{i=1}^n \left(e_i^{(k)}\right)^2}{\sum_{i=1}^n \left(y_i^{(k)} - \overline{y}^{(k)}\right)^2}, \qquad \overline{y}^{(k)} = \frac{1}{n}\sum_{i=1}^n y_i^{(k)}. \tag{68}$$

The remaining error measures are

$$\mathrm{RMSE}_k = \left(\frac{1}{n}\sum_{i=1}^n \left(e_i^{(k)}\right)^2\right)^{1/2}, \qquad \mathrm{MAE}_k = \frac{1}{n}\sum_{i=1}^n \left|e_i^{(k)}\right|, \tag{69}$$

$$\mathrm{MedAE}_k = \mathrm{median}\left(\left|e_1^{(k)}\right|, \ldots, \left|e_n^{(k)}\right|\right), \qquad \mathrm{MAPE}_k = \frac{100}{n}\sum_{i=1}^n \left|\frac{e_i^{(k)}}{\max\left(|y_i^{(k)}|, \varepsilon\right)}\right|, \tag{70}$$

$$\mathrm{MaxErr}_k = \max_{1\le i\le n}\left|e_i^{(k)}\right|, \qquad \mathrm{Bias}_k = \frac{1}{n}\sum_{i=1}^n e_i^{(k)}, \tag{71}$$

where ε is a small numerical guard used only to prevent division by zero. These metrics are computed per criterion and then averaged across criteria and walk-forward folds to obtain a stable summary of model performance. The interval layer is evaluated with empirical coverage, average width, and the Winkler score, while residual behavior is checked with the Durbin-Watson statistic,

$$DW = \frac{\sum_{i=2}^{n} (e_i - e_{i-1})^2}{\sum_{i=1}^{n} e_i^2}, \tag{72}$$

the Breusch-Pagan test for heteroscedasticity, and the Shapiro-Wilk test for residual normality. Validation follows the same walk-forward chronology used to train the meta-learner, so the reported numbers are genuinely out-of-sample summaries rather than resubstitution fits. This is the appropriate protocol for a 14-year panel in which temporal dependence, structural breaks, and heterogeneous missingness all matter to forecast quality.

3 Results

3.1 CRITIC Weighting

[Content placeholder.]

3.2 Historical Provincial Rankings (2011–2024)

[Content placeholder.]

3.3 Inter-Method Concordance and Robustness

[Content placeholder.]

3.4 Forecasting Model Performance

[Content placeholder.]

3.5 2025 Provincial Governance Forecasts and Prediction Intervals

[Content placeholder.]

3.6 Integrated MCDM Ranking of 2025 Forecasts

[Content placeholder.]

4 Discussion

4.1 Interpretation of CRITIC Weights

[Content placeholder.]

4.2 MCDM Method Convergence and Governance Interpretation

[Content placeholder.]

4.3 Forecasting Performance in Small-Panel Contexts

[Content placeholder.]

4.4 Conformal Prediction Interval Quality

[Content placeholder.]

4.5 Policy Implications for Vietnam's Governance Reform

[Content placeholder.]

4.6 Comparison with Related Approaches

[Content placeholder.]

4.7 Limitations

[Content placeholder.]

5 Conclusion

[Content placeholder.]

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